Abdulsalam Oluwaseun Ayeleru

AI Systems Engineer | Full-Stack Product Engineer | Mobile & Backend Architect

Mail | Portfolio | GitHub | LinkedIn | Blog

AI Systems and Product Engineer with deep experience architecting high-throughput backends, native mobile applications, and first-principles machine learning models. Built and deployed 3 production mobile apps across the Apple App Store and Google Play, managed 500+ production REST APIs, and authored research frameworks spanning multi-agent orchestration and computer vision.

Core Technical Skills

Languages & Backends: Python, TypeScript, JavaScript, SQL, NestJS, FastAPI, Node.js, Next.js, REST APIs, Webhooks, GraphQL
Databases & Systems: PostgreSQL (ACID transactions, relational indexing), MongoDB, DynamoDB, Pinecone, Docker, Linux, Git
AI & Computer Vision: PyTorch, TensorFlow, LLMs, Vision Transformers (ViT), U-Net, CNNs, YOLO, MediaPipe, TFLite
Cloud, Mobile & Stores: Apple App Store, Google Play Console, Capacitor, Firebase, AWS (Lambda, SQS, SAM, EventBridge), GitHub Actions CI/CD

Professional Experience

GoodTenants / Upward Ecosystem **March 2026 – Present**
Software Engineer & Product Engineer *Lagos, Nigeria (Hybrid)*

- Took complete ownership of launching **UpwardPay live on the Apple App Store**, managing guidelines compliance, TestFlight review cycles, and release management; shipped and operated **Upward Pay** and **Upward PM** on Google Play Store.
- Architected a high-throughput **monolithic NestJS server managing 500+ production REST APIs**, serving concurrent real-time requests across the Tenant app and Property Manager app while syncing with external property management platforms.
- Designed a complex relational PostgreSQL database schema with indexed constraints and deterministic transaction boundaries to eliminate race conditions and double-crediting during high-velocity rent collections and payouts.
- Engineered full-stack features using Next.js and Capacitor, achieving 90%+ code reuse across iOS, Android, and web with automated GitHub Actions CI/CD pipelines.

Codygo (Worknet) **May 2025 – December 2025**
AI Full-Stack & Serverless Engineer *Remote*

- Took complete engineering ownership of **sync-datasource**, an event-driven RAG synchronization pipeline on **AWS SAM, Lambda, SQS, EventBridge, and DynamoDB**, engineering cursor-based pagination across message queues to sync massive datasets beyond Lambda execution limits.
- Implemented multi-format document chunking, **SHA-256 content deduplication**, and automated **Pinecone vector indexing**, eliminating redundant embedding calls and reducing vector storage overhead.
- Engineered the **Worknet B2B embeddable AI Copilot widget** (`embed.js` and sandboxed cross-origin iframe runtime) served dynamically via AWS Lambda Function URLs with custom authentication and real-time proactive notification triggers.
- Built self-serve data source configuration screens and batch processing microservices in the admin console, enabling enterprise clients to connect knowledge sources (GitHub, Notion, files) and schedule automated cron sync cadences.

Tweakrr **March 2025 – May 2025**
AI Engineer & Backend Engineer *Remote*

- Developed document analysis pipelines using Python, FastAPI, and MongoDB to parse academic manuscripts, extract factual claims, and flag unsupported statements against scholarly literature.
- Engineered automated citation generation adhering to APA, MLA, IEEE, and Harvard referencing styles, reducing manual citation turnaround time by over 70%.

Institute of Advanced Medical Research and Training (IAMRAT / UCH) **December 2024 – March 2025**
AI Research & Systems Engineer *University College Hospital, Ibadan*

- Trained deep computer vision models with Google Path Foundation Vision Transformers and ResNet/MobileNet to classify breast cancer subtypes from whole-slide biopsy imagery.
- Constructed multimodal clinical decision-support workflows connecting vision predictions to Gemini diagnostic summaries; engineered full-stack FastAPI and React portals for pathologist trial annotation.

Selected Projects & Research Frameworks

Asera Core | *Open-Source Multi-Agent Supervisor & Long-Context Engine* **Python, Gemini 2.0, RLM**

- Architected a lightweight Python multi-agent supervisor framework implementing the MIT CSAIL Recursive Language Models (RLM) paradigm (arXiv:2512.24601), offloading unbounded documents to a sandboxed REPL to cut input tokens by 96.7%.
- Built parallel worker execution using asyncio.gather, reducing multi-agent delegation latency from 8.2s to 1.8s.

LoveDA Remote Sensing Segmentation | *Weakly-Supervised Aerial Vision* **PyTorch, U-Net, pCE Loss**

- Researched aerial semantic segmentation using sparse point clicks on the LoveDA Urban benchmark, discovering that 5% point density achieves peak generalization (+54.1% mIoU gain) while cutting manual polygon annotation time by over 95%.

RSNA Lumbar Spine 3D MRI Grading | *Volumetric Deep Learning Benchmark* **PyTorch, DICOM, GeM Pooling**

- Engineered a volumetric deep learning system grading 25 simultaneous structural targets across 5 disc levels from multi-sequence MRI scans, achieving 0.441 weighted log loss and 83.6% recall on high-risk severe spinal conditions.

Leadership, Mentorship & Education

- **Skillsphere & Techqings AI Mentor:** Mentored 200+ students across machine learning fundamentals, making AI practical and accessible while empowering women breaking into technology.
- **Larva Tech Academy:** Instructed and mentored 30+ aspiring developers in modern web architecture, TypeScript, and clean UI engineering.
- **RAIN-INN Robotics & AI Community Co-Lead (University of Ibadan):** Organized technical workshops and student AI/robotics cohorts.
- **Keynote Speaker:** Delivered practical software talks at the STUMPED AI Seminar and the 2026 Become Valuable Conference.

University of Ibadan | Bachelor of Science in Computer Science, Nigeria **Expected 2027**
Certifications: DeepLearning.AI Deep Learning Specialization | Hugging Face Transformers